

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Recalls a correct trig identity, which could lead to a correct answer	AO1.2	B1	(LHS $\equiv$) $\frac{\sin 2x}{1 + \tan^2 x}$
	Demonstrates a strategy for proving the identity, eg by converting all the terms on the LHS to cos and sin.	AO3.1a	M1	$\frac{2 \sin x \cos x}{1 + \tan^2 x}$ $\frac{2 \sin x \cos x}{\sec^2 x}$
	Concludes a rigorous mathematical argument to prove given identity AG	AO2.1	R1	$\equiv 2 \sin x \cos x \cos^2 x$ $\equiv 2 \sin x \cos^3 x$ ($\equiv$ RHS)
(b)	Uses identity to write integrand in the form $a \sin 2\theta \cos^3 2\theta$	AO1.1a	M1	$\int \frac{4 \sin 4\theta}{1 + \tan^2 2\theta} d\theta = \int 8 \sin 2\theta \cos^3 2\theta d\theta$ Let $u = \cos 2\theta$
	Correctly writes integrand as $8 \sin 2\theta \cos^3 2\theta$	AO1.1b	A1	then $\frac{du}{d\theta} = -2 \sin 2\theta \Rightarrow \sin 2\theta = -\frac{1}{2} \frac{du}{d\theta}$ $I = -4 \int u^3 \frac{du}{d\theta} d\theta$
	Selects an appropriate method for integrating, e.g. substitution $u = \cos 2\theta$, or by inspection PI by sight of $\cos^4 2\theta$	AO3.1a	M1	$= -4 \int u^3 du$ $= -u^4 + c$ $= -\cos^4 2\theta + c$
	Obtains $k \int u^3 du$ correctly PI by solution in form $k \cos^4 2\theta$, if by inspection	AO1.1a	M1	
	Obtains $-u^4$ or $-\cos^4 2\theta$ OE Only FT value of a	AO1.1b	A1F	
	Completes rigorous argument to obtain $-\cos^4 2\theta + c$ OE	AO2.1	R1	
				Total 9 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Recalls a correct trig identity, which could lead to a correct answer	AO1.2	B1	(LHS $\equiv$) $\cot^2 \theta - \cos^2 \theta$
Performs some correct algebraic manipulation and uses second identity to commence proof (at least two lines of argument)	AO2.1	R1	$\frac{\cos^2 \theta}{\sin^2 \theta} - \cos^2 \theta$ $\equiv \cos^2 \theta \left(\frac{1}{\sin^2 \theta} - 1 \right)$ $\equiv \cos^2 \theta (\operatorname{cosec}^2 \theta - 1)$ $\equiv \cos^2 \theta \cot^2 \theta$ ($\equiv$ RHS)
Concludes a rigorous mathematical argument to prove given identity AG Must start with one side and through clear logical steps arrive at the other side. In order to be sufficiently clear, each line should be a single step, unless clear further explanation is given.	AO2.1	R1	AG
Total 3 marks			

Q3.

(a)
$$\frac{\sec^2 x}{(\sec x + 1)(\sec x - 1)} = \frac{\sec^2 x}{\sec^2 x - 1}$$

$\sec^2 x = 1 + \tan^2 x$ used

*M1 for correct use of $\sec^2 x = 1 + \tan^2 x$ at least once or
($\operatorname{cosec}^2 x = 1 + \cot^2 x$)*

M1

$$= \frac{\sec^2 x}{\tan^2 x} \text{ or } \frac{1 + \tan^2 x}{\tan^2 x}$$

$$\left(= \frac{1}{\cos^2 x \tan^2 x} \right)$$

$$= \frac{1}{\sin^2 x} \text{ or } \cot^2 x + 1$$

Shown, with no errors

A1

$$= \operatorname{cosec}^2 x$$

AG (No errors, omissions or poor notations seen)

A1

3

- (b) $\operatorname{cosec}^2 x = \operatorname{cosec} x + 3$
 $\operatorname{cosec}^2 x - \operatorname{cosec} x - 3 = 0$
must have = 0
correct solution of the quadratic,
or by completing the square

B1

$$\operatorname{cosec} x = \frac{1 \pm \sqrt{13}}{2} \text{ or } (2.3... \text{ and } -1.3...)$$

$$\left(\operatorname{cosec} x = \pm \sqrt{\frac{13}{4} + \frac{1}{2}} \right)$$

PI by values for $\sin x$

M1

$$\sin x = \frac{2}{1 \pm \sqrt{13}}$$

B1F for $\operatorname{cosec} x = \frac{1}{\sin^2 x}$ seen or implied

B1F

$$= 0.434 \text{ and } -0.768 \text{ (or } -0.767)$$

PI

A1

B1 for any three values correct AWRT

©2025 Exam Papers Practice. All rights reserved.

B1

$$x = 26^\circ, 154^\circ, -50^\circ, -130^\circ$$

B1 for all four values correct AWRT and no extras
in the interval $-180 < x < 180^\circ$

B1

6

- (c) $2\theta - 60^\circ = x$
where x is a written value from candidate's (b) in degrees
PI by their answer

M1

$$\theta = 43^\circ, 5^\circ$$

CSO

A1

Ignore solutions outside interval $0^\circ < \theta < 90^\circ$

2

[11]

Q4.

$$\dots = 9 \sin^2 x + 6 \sin x \cos x + \cos^2 x +$$

$$\sin^2 x - 6 \sin x \cos x + 9 \cos^2 x$$

*Attempt at expanding both sets of brackets.
Minimum requirement either one of the two
expansions correct or 4 of these 6 terms seen.
Expanding and simplifying the given expression
in one step to get the correct two terms scores
this M1 and next A1*

M1

$$\dots = 10 \cos^2 x + 10 \sin^2 x$$

*Either correct pair of expansions and simplification
to remove $\sin x \cos x$ terms or full collecting of like
terms within the original correct expansion*

A1

$$= 10 (1 - \sin^2 x) + 10 \sin^2 x$$

*$\cos^2 x + \sin^2 x = 1$ clearly used. If identity is
applied correctly, but not directly, it must be
stated at the relevant point in the proof.*

M1

$$= 10 \text{ (which is an integer)}$$

*CSO [all previous 3 marks must have been
scored]*

*Condone absence of statement after 10
obtained correctly*

A1

[4]

Q5.

(a) (i) $R = 10$

$$R = 10$$

B1

$\tan \alpha = \frac{8}{6}, \quad \alpha = 53.1$

For α ; ft incorrect R

B1F

2

(ii) $\sin (2x + 53.1) = 0.7$

M1

$2x + 53.1 = 44.4$

one correct answer ; ft α and R

A1F

135.6 or 135.7 , 404.4 , 495.6 or 495.7

3 other correct answers – ignore extras

A1

$x = 41.2$ or 41.3 , 175.6 or 175.7 ,
 221.2 or 221.3 , 355.6 or 355.7

four solutions

CAO (with decimal place discrepancies)

Answers only: 0/4

A1

4

(b) (i) $\sin 2x = 2 \sin x \cos x$

EXAM PAPERS PRACTICE

B1

$\cos 2x = \cos^2 x - \sin^2 x$

identities for $\sin 2x$ and $\cos 2x$ in any correct form

B1

$$\frac{\sin 2x}{1 - \cos 2x} = \frac{2 \sin x \cos x}{1 - (1 - 2 \sin^2 x)} =$$

use of candidate's double angle formulae

M1

$$\frac{2 \sin x \cos x}{2 \sin^2 x} = \frac{\cos x}{\sin x} = \frac{1}{\tan x}$$

AG, CSO

A1

4

(ii) $\frac{1}{\tan x} = \tan x \quad \tan x = \pm 1$
*(see * below)*

M1A1

$x = 45,$

B1

135, 225, 315

if answers given without working, B1 max

$\frac{1}{\tan x} = \tan x$
if $\tan x$ seen and followed by correct answers without working 4 out of 4

A1

4

** If hence ignored, so working in sines and cosines, must simplify as far as:*

$\cos^2 x = \sin^2 x$	or	$\cos^2 x = \frac{1}{2}$	or	$\sin^2 x = \frac{1}{2}$	for M1
$\cos 2x = 0$	or	$\cos x = \pm \frac{1}{\sqrt{2}}$	or	$\sin x = \pm \frac{1}{\sqrt{2}}$	for A1

[14]

Q6.

(a) $\sin(x + 2x) = \sin x \cos 2x + \cos x \sin 2x$

©2025 Exam Papers Practice. All rights reserved.

M1

$= \sin x(1 - 2 \sin^2 x) + \cos x(2 \sin x \cos x)$

double angles; ACF ISW condone missing x

B1B1

$= \sin x(1 - 2 \sin^2 x) + 2 \sin x(1 - \sin^2 x)$

all in sin x, correct expression

A1

$= 3 \sin x - 2 \sin^3 x - 2 \sin^3 x$

$= 3 \sin x - 4 \sin^3 x$

CSO AG

A1

5

(b) $\sin^3 x = a \sin x + b \sin 3x$

attempt to solve for $\sin^3 x$ where $a \neq 0$ and $b \neq 0$

M1

$$\int \sin^3 x \, dx = -a \cos x - \frac{b}{3} \cos 3x$$

either integral correct

F on a, b

A1F

$$\int \sin^3 x \, dx = \frac{1}{4} \left(-3 \cos x + \frac{1}{3} \cos 3x \right) (+C)$$

CAO

alternative method by parts (see end of mark scheme)

A1

ALTERNATIVE METHOD 1

$$\int \sin^3 x \, dx = \int \sin^2 x \sin x \, dx$$

identify parts and attempt to integrate

M1

$$= -\sin^2 x \cos x - \int -2 \cos x \sin x \cos x \, dx$$

$$= -\sin^2 x \cos x - \frac{2}{3} \cos^3 x (+C)$$

©2025 Exam Papers Practice. All rights reserved.

A2

ALTERNATIVE METHOD 2

$$\int \sin^3 x \, dx = \int \sin^2 x \, d(-\cos x)$$

condone sign error

M1

$$= \int -(1 - \cos^2 x) \, d(\cos x)$$

$$= -\cos x + \frac{1}{3} \cos^3 x (+C)$$

A2

ALTERNATIVE METHOD 3

$$\int \sin x \sin^2 x dx$$

$$\int \sin x (1 - \cos^2 x) dx$$

this form and attempt to integrate

M1

$$= -\cos x + \frac{1}{3} \cos^3 x (+C)$$

A2

3

[8]

Q7.

(a) $\cos 2x = 1 - 2\sin^2 x$

B1

1

(b) (i) $3\sin x - \cos 2x = 3\sin x - (1 - 2\sin^2 x)$

Candidate's cos 2x or sin² x

M1

$$= 3\sin x - 1 + 2\sin^2 x$$

AG

A1

2

(ii) $2\sin^2 x + 3\sin x - 2 = 0$

Soluble quadratic form

M1

$$(2\sin x - 1)(\sin x + 2) = 0$$

Attempt to solve (allow one error in formula, allow sign errors)

M1

$$\sin x = \frac{1}{2} \quad x = 30 \quad x = 150$$

M1

*sin⁻¹ and two solutions (0° < x < 360°)
A0 if radians*

A1

4

Allow misread for $2\sin^2 x + 3\sin x - 1 = 0$

Soluble quadratic form

(M1)

$$\sin x = \frac{-3 \pm \sqrt{17}}{4}$$

Use of formula (allow one error)

(M1)

$$x = 16.3^\circ, 163.7^\circ$$

Max 3/4

(A1)

(c) $\int \frac{1}{2}(1 - \cos 2x) = \frac{x}{2} - \frac{\sin 2x}{4} (+c)$

*M1 – solve integral, must have 2 terms
for $\sin^2 x$ from (a)*

M1A1

2

[9]

Q8.

(a) $\tan(x+x) = \frac{\tan x + \tan x}{1 - \tan x \tan x} \left(= \frac{2 \tan x}{1 - \tan^2 x} \right)$

A = B = x used

M1

A1

2

(b) $2 - 2 \tan x - \frac{2 \tan x (1 - \tan^2 x)}{2 \tan x}$

Substitute from (a)

M1

$$2 - 2 \tan x - (1 - \tan x)(1 + \tan x)$$

Simplification $2 - 2 \tan x - (1 - \tan^2 x)$

M1

$$(1 - \tan x)(2 - (1 + \tan x))$$

$$2 - 2 \tan x - 1 + \tan^2 x$$

M1

$$(1 - \tan x)^2$$

AG (convincingly obtained)

$$= (\tan x - 1)^2 = (1 - \tan x)^2$$

Any equivalent method

A1

4

[6]

Q9.

(a) $\tan x (+ c)$

B1

1

(b) $f(x) = \frac{\cos x}{\sin x}$

$$f'(x) = \frac{-\sin^2 x - \cos^2 x}{\sin^2 x}$$

$\frac{\pm \sin^2 x \pm \cos^2 x}{\sin^2 x}$
quotient rule

M1

A1

EXAM PAPERS PRACTICE

$$= \frac{-1}{\sin^2 x}$$

©2025 Exam Papers Practice. All rights reserved.

use of $\sin^2 x + \cos^2 x = 1$

A1

$$= -\operatorname{cosec}^2 x$$

AG CSO

A1

4

Special cases

$$f(x) = \frac{\cot x}{1}$$

$$f'(x) = \frac{1 \times -\operatorname{cosec}^2 x - \cot x \times 0}{1^2}$$

M1

$$= -\operatorname{cosec}^2 x$$

A1
(max 2/4)

Or

$$f(x) = \frac{1}{\tan x}$$

$$f'(x) = \frac{\tan x \times 0 - 1 \times \sec^2 x}{\tan^2 x}$$

M1A1

$$= \frac{-\sec^2 x}{\tan^2 x}$$

$$= \frac{-1}{\sin^2 x} = -\operatorname{cosec}^2$$

A1
(max 3/4)

(c) LHS = $\tan^2 x + \cot^2 x + 2 \tan x \cot x$
expanding

M1

$$= \tan^2 x + 1 + \cot^2 x + 1$$

correct use of trig identities

M1

$$= \sec^2 x + \operatorname{cosec}^2 x$$

CSO

A1

3

$$= \text{RHS}$$

(d) $\int (\tan x + \cot x)^2 dx = \int \sec^2 x + \operatorname{cosec}^2 x dx$

use of identity

M1

$$= [\tan x - \cot x]_{0.5}^1$$

$\pm \tan x \pm \cot x$ OE

M1
A1

$$= 0.9153 - -1.2842$$

$$= 2.2$$

AWRT

A1

4

[12]

Q10.

- (a) (i) Stretch **(I)** in y -direction **(II)**

Scale factor 2 **(III)**

>1 transformation is M0.

M1 for **(I)** and either **(II)** or **(III)** or **(III)**

M1A1

2

- (ii) Reflection;

M1

in x -axis

'Reflection'/'reflect(ed)'

(or in y -axis or $y = 0$ or $x = 0$)

A1

2

- (iii) Translation;

'Translation'/'translate (d)'

B1

$$\begin{bmatrix} 30^\circ \\ 0 \end{bmatrix}$$

Accept **full** equivalent in words provided linked to 'translation/move/shift' and **positive** x -direction
(Note: B0 B1 is possible)

B1

2

- (b) $\{\theta - 30^\circ = \sin^{-1}(0.7) = 44.4\dots^\circ$

Inverse sine of 0.7 PI eg by sight of 44, 74 or better

M1

..... = $180^\circ - 44.4^\circ$

Valid method for 2nd angle

m1

$\theta = 74.4^\circ, 165.6^\circ$

Condone >1 dp accuracy

A1

3

(c) $\dots = \cos^2 x + 2\cos x \sin x + \sin^2 x + \cos^2 x - 2\cos x \sin x + \sin^2 x$

Award for either bracket expanded correctly

M1

$$\dots = 2\cos^2 x + 2\sin^2 x$$

OE

A1

$$= 2(\cos^2 x + \sin^2 x) = 2(1)$$

$\cos^2 x + \sin^2 x = 1$ stated or used.

M1

$$= 2$$

AG (be convinced)

A1

4

[13]

Q11.

(a) (i) $\sin 2x = 2 \sin x \cos x$

B1

1

(ii) $\cos 2x = 2 \cos^2 x - 1$

B1

1

(b) $\sin 2x - \tan x = 2 \sin x \cos x - \frac{\sin x}{\cos x}$

Use of their $\cos 2x$ or $\sin 2x$

M1

$$= \sin x \left(2 \cos x - \frac{1}{\cos x} \right)$$

Use of $\tan x = \frac{\sin x}{\cos x}$ and the other double angle identity

M1

$$= \sin x \left(\frac{2 \cos^2 x - 1}{\cos x} \right) = \tan x \cos 2x$$

AG convincingly obtained

A1

3

(c) $\tan x \cos 2x = 0 \quad x = 180$

Ignore $x = 0$, $x = 360^\circ$ & any others outside range

B1

$$\cos 2x = 0 \text{ or } \cos^2 x = \frac{1}{2} \left(\text{or } \sin^2 x = \frac{1}{2} \right)$$

M1

$$x = 45$$

A1

$$x = 135, 225, 315$$

CAO max 3/4 for answers in radians

B1

4

[9]